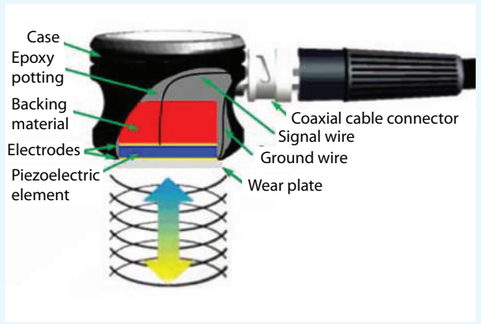


Figure TF12-2 The ultrasonic transcducer uses piezoelectric crystals.

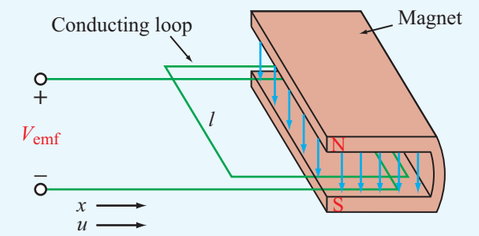


Figure TF12-3 In a Faraday accelerometer, the induced emf is directly proportional to the velocity of the loop (into and out of the magnet's cavity).

magnetic flux passing through the loop. For the configuration in **Fig. TF12-3**,

$$V_{\text{emf}} = -uB_0l,$$

where $u = dx/dt$ is the ***velocity*** of the loop (into or out of the magnet's cavity) with the direction of u defined as positive when the loop is moving inward into the cavity, B_0 is the magnetic field of the magnet, and l is the loop width. With B_0 and l being constant, the variation of $V_{\text{emf}}(t)$ with time t becomes a direct indicator of the time variation of $u(t)$. The time derivative of $u(t)$ provides the ***acceleration*** $a(t)$.

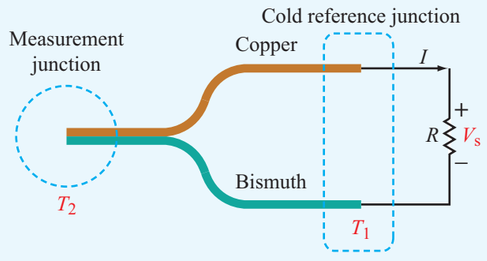


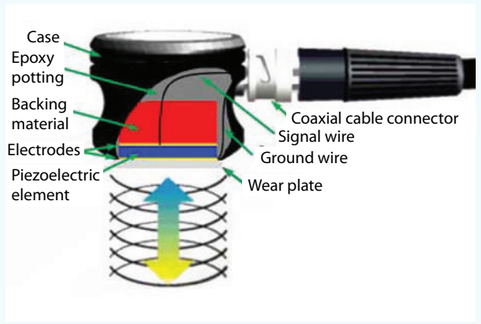
Figure TF12-4 Principle of the thermocouple.

Thermocouple

In 1821, ***Thomas Seebeck*** discovered that when a junction made of two different conducting materials, such as bismuth and copper, is heated it generates a thermally induced emf, which we now call the ***Seebeck potential*** V_S (**Fig. TF12-4**). When connected to a resistor, a current given by $I = V_S/R$ flows through the resistor.

This feature was advanced by ***A. C. Becquerel*** in 1826 as a means to measure the unknown temperature T_2 of a junction relative to a temperature T_1 of a (cold) reference junction. Today, such a generator of ***thermoelectricity*** is called a ***thermocouple***. Initially, an ice bath was used to maintain T_1 at 0°C, but in today's temperature sensor designs, an artificial cold junction is used instead. The artificial junction is an electric circuit that generates a potential equal to that expected from a reference junction at temperature T_1.

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图TF12-2 超声波换能器采用压电晶体。

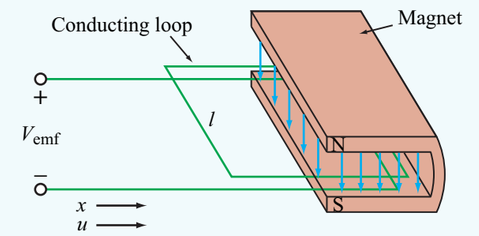
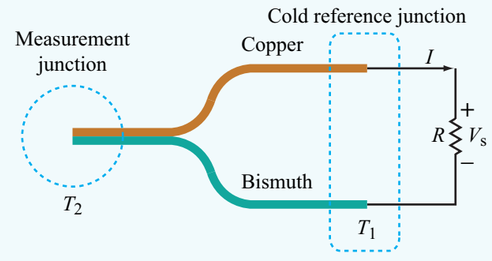


图 TF12-3 在法拉第加速度计中，感应电动势与回路速度（进出磁体腔）成正比。

通过环路的磁通量。对于图 TF12-3 中的配置，

$$V_{\text{emf}} = -uB_0l,$$

其中 u = dx/dt 是环路的速度（进入或离开磁体空腔），当环路向内移动到空腔中时，u 的方向定义为正，B_0 是磁体的磁场，l 是环路宽度。当 B_0 和 l 恒定时，$V_{\text{emf}}(t)$ 随时间 t 的变化就成为 u(t) 时间变化的直接指标。 u(t) 的时间导数提供加速度 a(t)。



图TF12-4 热电偶原理。

热电偶

1821 年，托马斯·塞贝克 (Thomas Seebeck) 发现，当由两种不同的导电材料（例如铋和铜）制成的结被加热时，会产生热感应电动势，我们现在将其称为塞贝克电势 V_S (图 1。 TF12-4)。当连接到电阻器时，由 $I = V_S/R$ 给出的电流流过电阻器。

这一功能由 A. C. Becquerel 于 1826 年提出，作为一种测量结点相对于（冷）参考结点温度 T_1 的未知温度 T_2 的方法。如今，这种热电发生器被称为热电偶。最初，使用冰浴将 T_1 维持在 0°C，但在当今的温度传感器设计中，改用人工冷端。人工结是一种电路，它产生的电势等于温度 T_1 下参考结预期的电势。

Exercise 6-6: Determine (a) the relaxation time constant and (b) the time it takes for a charge density to decay to 1% of its initial value in quartz given that $\varepsilon_r = 5$ and $\sigma = 10^{-17}$ S/m.

Answer: (a) $\tau_r = 51.2$ days, (b) 236 days. (See EM.)

6-11 Electromagnetic Potentials

Our discussion of Faraday's and Ampère's laws revealed two aspects of the link between time-varying electric and magnetic fields. We now examine the implications of this interconnection on the electric scalar potential V and the vector magnetic potential $\mathbf{A}$.

In the static case, Faraday's law reduces to

$$\nabla \times \mathbf{E} = 0, \qquad \textbf{(static case)} \tag{6.62}$$

which states that the electrostatic field $\mathbf{E}$ is conservative. According to the rules of vector calculus, if a vector field $\mathbf{E}$ is conservative, it can be expressed as the gradient of a scalar. Hence, in Chapter 4, we defined $\mathbf{E}$ as

$$\mathbf{E} = -\nabla V. \qquad \textbf{(electrostatics)} \tag{6.63}$$

In the dynamic case, Faraday's law is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{6.64}$$

In view of the relation $\mathbf{B} = \nabla \times \mathbf{A}$, Eq. (6.64) can be expressed as

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}), \tag{6.65}$$

which can be rewritten as

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right) = 0. \qquad \textbf{(dynamic case)} \tag{6.66}$$

Let us for the moment define

$$\mathbf{E}' = \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}. \tag{6.67}$$

Using this definition, Eq. (6.66) becomes

$$\nabla \times \mathbf{E}' = 0. \tag{6.68}$$

Following the same logic that led to Eq. (6.63) from Eq. (6.62), we define

$$\mathbf{E}' = -\nabla V. \tag{6.69}$$

Upon substituting Eq. (6.67) for $\mathbf{E}'$ in Eq. (6.69) and then solving for $\mathbf{E}$, we have

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}. \qquad \textbf{(dynamic case)} \tag{6.70}$$

Equation (6.70) reduces to Eq. (6.63) in the static case.

When the scalar potential V and the vector potential $\mathbf{A}$ are known, $\mathbf{E}$ can be obtained from Eq. (6.70), and $\mathbf{B}$ can be obtained from

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{6.71}$$

Next we examine the relations between the potentials, V and $\mathbf{A}$, and their sources: the charge and current distributions ρ_v and $\mathbf{J}$ in the time-varying case.

6-11.1 Retarded Potentials

Consider the situation depicted in Fig. 6-16. A charge distribution ρ_v exists over a volume $\mathcal{V}'$ embedded in a perfect dielectric with permittivity ε. Were this a static charge distribution, then from Eq. (4.58a) the electric potential $V(\mathbf{R})$ at an observation point in space specified by the position vector $\mathbf{R}$ would be

$$V(\mathbf{R}) = \frac{1}{4\pi\varepsilon}\int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R}_i)}{R'}\, d\mathcal{V}', \tag{6.72}$$

where $\mathbf{R}_i$ denotes the position vector of an elemental volume $\Delta\mathcal{V}'$ containing charge density $\rho_v(\mathbf{R}_i)$, and $R' = |\mathbf{R} - \mathbf{R}_i|$ is the distance between $\Delta\mathcal{V}'$ and the observation point. If the charge

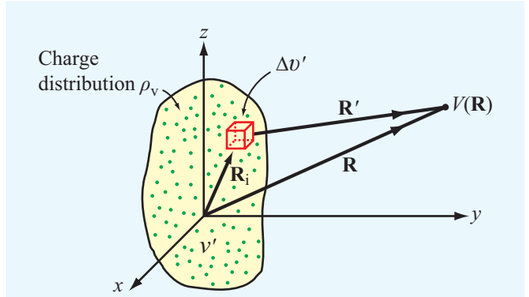


Figure 6-16 Electric potential $V(\mathbf{R})$ due to a charge distribution ρ_v over a volume $\mathcal{V}'$.

练习 6-6：在 $\varepsilon_r = 5$ 和 $\sigma = 10^{-17}$ S/m 的情况下，确定 (a) 弛豫时间常数和 (b) 石英中电荷密度衰减到其初始值的 1% 所需的时间。

答案：(a) $\tau_r = 51.2$ 天，(b) 236 天。（参见EM。）

6-11 电磁势

对法拉第定律和安培–麦克斯韦定律的讨论，揭示了时变电场与磁场相互联系的两个方面。下面研究这种联系对电标量势 V 和磁矢势 A 的影响。

在静态情况下，法拉第定律简化为

$$\nabla \times \mathbf{E} = 0, \qquad \textbf{(static case)} \tag{6.62}$$

这说明静电场 E 是保守的。根据向量微积分的规则，如果向量场E是保守的，它可以表示为标量的梯度。因此，在第 4 章中，我们将 E 定义为

$$\mathbf{E} = -\nabla V. \qquad \textbf{(electrostatics)} \tag{6.63}$$

在动态情况下，法拉第定律为

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{6.64}$$

鉴于关系 $\mathrm{B} = \nabla \times \mathrm{A}$，等式： (6.64) 可以表示为

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}), \tag{6.65}$$

可以重写为

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right) = 0. \qquad \textbf{(dynamic case)} \tag{6.66}$$

让我们暂时定义

$$\mathbf{E}' = \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}. \tag{6.67}$$

使用这个定义，方程。 (6.66) 变为

$$\nabla \times \mathbf{E}' = 0. \tag{6.68}$$

遵循与方程相同的逻辑。 (6.63) 来自方程式（6.62），我们定义

$$\mathbf{E}' = -\nabla V. \tag{6.69}$$

代入方程式后。 (6.67) 对于方程中的 E′ (6.69) 然后求解 E，我们有

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}. \qquad \textbf{(dynamic case)} \tag{6.70}$$

方程（6.70）简化为方程。 (6.63) 在静态情况下。

当标量势V和矢量势A已知时，可由式(1)求出E。 (6.70)，B 可由下式获得

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{6.71}$$

接下来，我们检查电势 V 和 A 及其来源之间的关系：时变情况下的电荷和电流分布 ρ_v 和 J。

6-11.1 迟滞电位

考虑图 6-16 中描述的情况。电荷分布 ρ_v 存在于嵌入介电常数为 ε 的理想电介质中的体积 $\mathcal{V}'$ 上。如果这是一个静电荷分布，则可以从方程式得出： (4.58a) 由位置向量 R 指定的空间观察点处的电势 V (R) 为

$$V(\mathbf{R}) = \frac{1}{4\pi\varepsilon}\int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R}_i)}{R'}\, d\mathcal{V}', \tag{6.72}$$

其中，$\mathrm{R_i}$ 表示包含电荷密度 $\rho_v(\mathrm{R_i})$ 的单元体积 $\Delta\mathcal{V}'$ 的位置向量，$\mathrm{R}' = |\mathrm{R} - \mathrm{R_i}|$ 是 $\Delta\mathcal{V}'$ 与观测点之间的距离。如果电荷

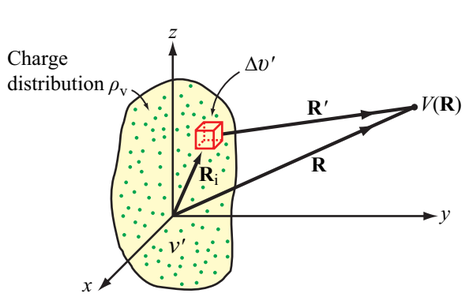


图 6-16 体积 $\mathcal{V}'$ 上的电荷分布 ρ_v 产生的电势 V (R)。

distribution is time-varying, we may be tempted to rewrite Eq. (6.72) for the dynamic case as

$$V(\mathbf{R},t) = \frac{1}{4\pi\varepsilon}\int_{v'}\frac{\rho_v(\mathbf{R}_i,t)}{R'}\,dv', \tag{6.73}$$

but such a form does not account for "reaction time." If V_1 is the potential due to a certain distribution ρ_{v1} and ρ_{v1} were to suddenly change to ρ_{v2}, it will take a finite amount of time before V_1 a distance R' away changes to V_2. *In other words, $V(\mathbf{R},t)$ cannot change instantaneously.* The delay time is equal to $t' = R'/u_p$, where u_p is the velocity of propagation in the medium between the charge distribution and the observation point. Thus, $V(\mathbf{R},t)$ at time t corresponds to ρ_v at an earlier time, that is, $(t-t')$. Hence, Eq. (6.73) should be rewritten as

$$V(\mathbf{R},t) = \frac{1}{4\pi\varepsilon}\int_{v'}\frac{\rho_v(\mathbf{R}_i,\,t-R'/u_p)}{R'}\,dv' \quad \text{(V)}, \tag{6.74}$$

and $V(\mathbf{R},t)$ is appropriately called the ***retarded scalar potential***. If the propagation medium is vacuum, u_p is equal to the velocity of light c.

Similarly, the ***retarded vector potential*** $\mathbf{A}(\mathbf{R},t)$ is related to the distribution of current density $\mathbf{J}$ by

$$\mathbf{A}(\mathbf{R},t) = \frac{\mu}{4\pi}\int_{v'}\frac{\mathbf{J}(\mathbf{R}_i,\,t-R'/u_p)}{R'}\,dv' \quad \text{(Wb/m)}. \tag{6.75}$$

This expression is obtained by extending the expression for the magnetostatic vector potential $\mathbf{A}(\mathbf{R})$ given by Eq. (5.65) to the time-varying case.

6-11.2 Time-Harmonic Potentials

The expressions given by Eqs. (6.74) and (6.75) for the retarded scalar and vector potentials are valid under both static and dynamic conditions and for any type of time dependence of the source functions ρ_v and $\mathbf{J}$. Because V and $\mathbf{A}$ depend linearly on ρ_v and $\mathbf{J}$, and as $\mathbf{E}$ and $\mathbf{B}$ depend linearly on V and $\mathbf{A}$, the relationships interconnecting all of these quantities obey the rules of linear systems. When analyzing linear systems, we can take advantage of sinusoidal-time functions to determine the system's response to a source with arbitrary time dependence. As was noted in Section 1-7, if the time dependence is described by a (nonsinusoidal) periodic time function, it can always be expanded into a Fourier series of sinusoidal components, and if the time function is nonperiodic, it can be represented by a Fourier integral. In either case, if the response of the linear system is known for all steady-state sinusoidal excitations, the principle of superposition can be used to determine its response to an excitation with arbitrary time dependence. Thus, the sinusoidal response of the system constitutes a fundamental building block that can be used to determine the response due to a source described by an arbitrary function of time. The term ***time-harmonic*** is often used in this context as a synonym for "steady-state sinusoidal time-dependent."

In this subsection, we derive expressions for the scalar and vector potentials due to time-harmonic sources. Suppose that $\rho_v(\mathbf{R}_i,t)$ is a sinusoidal-time function with angular frequency ω given by

$$\rho_v(\mathbf{R}_i,t) = \rho_v(\mathbf{R}_i)\cos(\omega t+\phi). \tag{6.76}$$

Phasor analysis, which was first introduced in Section 1-7 and then used extensively in Chapter 2 to study wave propagation on transmission lines, is a useful tool for analyzing time-harmonic scenarios. A time harmonic charge distribution $\rho_v(\mathbf{R}_i,t)$ is related to its phasor $\tilde{\rho}_v(\mathbf{R}_i)$ as

$$\rho_v(\mathbf{R}_i,t) = \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{j\omega t}\right], \tag{6.77}$$

Comparison of Eqs. (6.76) and (6.77) shows that in the present case $\tilde{\rho}_v(\mathbf{R}_i) = \rho_v(\mathbf{R}_i)\,e^{j\phi}$.

Next, we express the retarded charge density $\rho_v(\mathbf{R}_i,t-R'/u_p)$ in phasor form by replacing t with $(t-R'/u_p)$ in Eq. (6.77):

$$\begin{aligned}\rho_v(\mathbf{R}_i,t-R'/u_p) &= \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{j\omega(t-R'/u_p)}\right]\\ &= \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{-j\omega R'/u_p}e^{j\omega t}\right]\\ &= \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{-jkR'}e^{j\omega t}\right],\end{aligned} \tag{6.78}$$

where

$$k = \frac{\omega}{u_p} \tag{6.79}$$

is called the ***wavenumber*** or phase constant of the propagation medium. (In general, the phase constant is denoted by the symbol "β", but for lossless dielectric media, it is commonly denoted by the symbol "k" and called the wavenumber.) Similarly, we define the phasor $\tilde{V}(\mathbf{R})$ of the time function $V(\mathbf{R},t)$ according to

$$V(\mathbf{R},t) = \mathfrak{Re}\left[\tilde{V}(\mathbf{R})\,e^{j\omega t}\right]. \tag{6.80}$$

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分布是随时间变化的，我们可能会想重写方程。 (6.72) 对于动态情况为

$$V(\mathbf{R},t) = \frac{1}{4\pi\varepsilon}\int_{v'}\frac{\rho_v(\mathbf{R}_i,t)}{R'}\,dv', \tag{6.73}$$

但这种形式并没有考虑“反应时间”。如果 V_1 是某个分布 ρ_{v1} 产生的势，并且 ρ_{v1} 突然变为 ρ_{v2}，则距离 R′ 远的 V_1 变为 V_2 需要有限的时间。换句话说，V(R,t)不能瞬时改变。延迟时间等于 $t' = R'/u_p$，其中 u_p 是电荷分布和观察点之间的介质中的传播速度。因此，时间t处的V(R,t)对应于较早时间处的ρ_v，即(t−t′)。因此，等式。 (6.73) 应重写为

$$V(\mathbf{R},t) = \frac{1}{4\pi\varepsilon}\int_{v'}\frac{\rho_v(\mathbf{R}_i,\,t-R'/u_p)}{R'}\,dv' \quad \text{(V)}, \tag{6.74}$$

V (R,t) 被适当地称为延迟标量势。如果传播介质是真空，u_p等于光速c。

类似地，延迟矢量势 A(R,t) 与电流密度 J 的分布相关：

$$\mathbf{A}(\mathbf{R},t) = \frac{\mu}{4\pi}\int_{v'}\frac{\mathbf{J}(\mathbf{R}_i,\,t-R'/u_p)}{R'}\,dv' \quad \text{(Wb/m)}. \tag{6.75}$$

该表达式是通过扩展方程 1 给出的静磁矢量势 A(R) 的表达式而获得的。 (5.65) 到时变的情况。

6-11.2 时谐势

等式给出的表达式。对于延迟标量势和矢量势，方程 (6.74) 和 (6.75) 在静态和动态条件下以及对于源函数 ρ_v 和 J 的任何类型的时间依赖性都有效。因为 V 和 A 线性依赖于 ρ_v 和 J，并且 E 和 B 线性依赖于 V 和 A，所以所有这些量之间的相互关系遵循线性系统的规则。在分析线性系统时，我们可以利用正弦时间函数来确定系统对具有任意时间依赖性的源的响应。如第 1-7 节所述，如果时间依赖性由（非正弦）周期时间函数描述，则它始终可以展开为正弦分量的傅立叶级数，并且如果时间函数是非周期的，它可以用傅里叶积分来表示。在任一情况下，如果线性系统的响应对于所有稳态正弦激励都是已知的，则可以使用叠加原理来确定其对具有任意时间依赖性的激励的响应。因此，系统的正弦响应构成了一个基本构建块，可用于确定由任意时间函数描述的源引起的响应。在这种情况下，术语“时谐”通常用作“稳态正弦时间相关”的同义词。

在本小节中，我们推导了时谐源引起的标量势和矢量势的表达式。假设 $\rho_v(R_i,t)$ 是正弦时间函数，角频率 ω 为

$$\rho_v(\mathbf{R}_i,t) = \rho_v(\mathbf{R}_i)\cos(\omega t+\phi). \tag{6.76}$$

相量分析首先在第 1-7 节中介绍，然后在第 2 章中广泛用于研究传输线上的波传播，它是分析时谐波场景的有用工具。时间谐波电荷分布 $\rho_v(R_i,t)$ 与其相量 $\tilde{\rho}_v(R_i)$ 相关，如下所示

$$\rho_v(\mathbf{R}_i,t) = \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{j\omega t}\right], \tag{6.77}$$

等式的比较(6.76) 和 (6.77) 表明在本例中$\tilde{\rho}_v(R_i) = \rho_v(R_i)\,e^{j\phi}$。

接下来，我们将等式中的 t 替换为 $(t - R'/u_p)$，以相量形式表示延迟电荷密度 $\rho_v(R_i,t - R'/u_p)$。 (6.77)：

$$\begin{aligned}\rho_v(\mathbf{R}_i,t-R'/u_p) &= \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{j\omega(t-R'/u_p)}\right]\\ &= \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{-j\omega R'/u_p}e^{j\omega t}\right]\\ &= \mathfrak{Re}\left[\tilde{\rho}_v(\mathbf{R}_i)\,e^{-jkR'}e^{j\omega t}\right],\end{aligned} \tag{6.78}$$

在哪里

$$k = \frac{\omega}{u_p} \tag{6.79}$$

称为传播介质的波数或相位常数。 （一般情况下，相位常数用符号“β”表示，但对于无损电介质，通常用符号“k”表示，称为波数。）类似地，我们根据下式定义时间函数V（R，t）的相量~V（R）：

$$V(\mathbf{R},t) = \mathfrak{Re}\left[\tilde{V}(\mathbf{R})\,e^{j\omega t}\right]. \tag{6.80}$$

Using Eqs. (6.78) and (6.80) in Eq. (6.74) gives

$$\mathfrak{Re}\left[\widetilde{V}(\mathbf{R})\,e^{j\omega t}\right] = \mathfrak{Re}\left[\frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\tilde{\rho}_{\mathrm{v}}(\mathbf{R}_{\mathrm{i}})\,e^{-jkR'}}{R'}\,e^{j\omega t}\,d\upsilon'\right]. \tag{6.81}$$

By equating the quantities inside the square brackets on both sides of Eq. (6.81) and canceling the common $e^{j\omega t}$ factor, we obtain the phasor-domain expression

$$\widetilde{V}(\mathbf{R}) = \frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\tilde{\rho}_{\mathrm{v}}(\mathbf{R}_{\mathrm{i}})\,e^{-jkR'}}{R'}\,d\upsilon' \quad (\mathrm{V}). \tag{6.82}$$

For any given charge distribution, Eq. (6.82) can be used to compute $\widetilde{V}(\mathbf{R})$. Then the resultant expression can be used in Eq. (6.80) to find $V(\mathbf{R},t)$. Similarly, the expression for $\mathbf{A}(\mathbf{R},t)$ given by Eq. (6.75) can be transformed into

$$\mathbf{A}(\mathbf{R},t) = \mathfrak{Re}\left[\widetilde{\mathbf{A}}(\mathbf{R})\,e^{j\omega t}\right] \tag{6.83}$$

with

$$\widetilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi}\int_{\upsilon'}\frac{\widetilde{\mathbf{J}}(\mathbf{R}_{\mathrm{i}})\,e^{-jkR'}}{R'}\,d\upsilon', \tag{6.84}$$

where $\widetilde{\mathbf{J}}(\mathbf{R}_{\mathrm{i}})$ is the phasor function corresponding to $\mathbf{J}(\mathbf{R}_{\mathrm{i}},t)$.

The magnetic field phasor $\widetilde{\mathbf{H}}$ corresponding to $\widetilde{\mathbf{A}}$ is given by

$$\widetilde{\mathbf{H}} = \frac{1}{\mu}\nabla\times\widetilde{\mathbf{A}}. \tag{6.85}$$

Recall that differentiation in the time domain is equivalent to multiplication by $j\omega$ in the phasor domain, and in a nonconducting medium ($\mathbf{J}=0$), so Ampère's law given by Eq. (6.41) becomes

$$\nabla\times\widetilde{\mathbf{H}} = j\omega\varepsilon\widetilde{\mathbf{E}} \quad \text{or} \quad \widetilde{\mathbf{E}} = \frac{1}{j\omega\varepsilon}\nabla\times\widetilde{\mathbf{H}}. \tag{6.86}$$

Hence, given a time-harmonic, current-density distribution with phasor $\widetilde{\mathbf{J}}$, Eqs. (6.84) to (6.86) can be used successively to determine both $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$. The phasor vectors $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ also are related by the phasor form of Faraday's law:

$$\nabla\times\widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}} \quad \text{or} \quad \widetilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla\times\widetilde{\mathbf{E}}. \tag{6.87}$$

Example 6-8: Relating E to H

In a nonconducting medium with $\varepsilon = 16\varepsilon_0$ and $\mu = \mu_0$, the electric field intensity of an electromagnetic wave is

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,10\sin(10^{10}t - kz) \quad (\mathrm{V/m}). \tag{6.88}$$

Determine the associated magnetic field intensity $\mathbf{H}$ and find the value of k.

Solution: We begin by finding the phasor $\widetilde{\mathbf{E}}(z)$ of $\mathbf{E}(z,t)$. Since $\mathbf{E}(z,t)$ is given as a sine function and phasors are defined in this book with reference to the cosine function, we rewrite Eq. (6.88) as

$$\begin{aligned}\mathbf{E}(z,t) &= \hat{\mathbf{x}}\,10\cos(10^{10}t - kz - \pi/2) \quad (\mathrm{V/m})\\ &= \mathfrak{Re}\left[\widetilde{\mathbf{E}}(z)\,e^{j\omega t}\right],\end{aligned} \tag{6.89}$$

with $\omega = 10^{10}$ (rad/s) and

$$\begin{aligned}\widetilde{\mathbf{E}}(z) &= \hat{\mathbf{x}}\,10e^{-jkz}e^{-j\pi/2}\\ &= -\hat{\mathbf{x}}j10e^{-jkz}.\end{aligned} \tag{6.90}$$

To find both $\widetilde{\mathbf{H}}(z)$ and k, we will perform a "circle": We will use the given expression for $\widetilde{\mathbf{E}}(z)$ in Faraday's law to find $\widetilde{\mathbf{H}}(z)$; then we will use $\widetilde{\mathbf{H}}(z)$ in Ampère's law to find $\widetilde{\mathbf{E}}(z)$, which we will then compare with the original expression for $\widetilde{\mathbf{E}}(z)$; and the comparison will yield the value of k. Application of Eq. (6.87) gives

$$\begin{aligned}\widetilde{\mathbf{H}}(z) &= -\frac{1}{j\omega\mu}\nabla\times\widetilde{\mathbf{E}}\\ &= -\frac{1}{j\omega\mu}\begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ -j10e^{-jkz} & 0 & 0\end{vmatrix}\\ &= -\frac{1}{j\omega\mu}\left[\hat{\mathbf{y}}\,\frac{\partial}{\partial z}(-j10e^{-jkz})\right]\\ &= -\hat{\mathbf{y}}j\,\frac{10k}{\omega\mu}e^{-jkz}.\end{aligned} \tag{6.91}$$

使用方程式。式（6.78）和（6.80）(6.74) 给出

$$\mathfrak{Re}\left[\widetilde{V}(\mathbf{R})\,e^{j\omega t}\right] = \mathfrak{Re}\left[\frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\tilde{\rho}_{\mathrm{v}}(\mathbf{R}_{\mathrm{i}})\,e^{-jkR'}}{R'}\,e^{j\omega t}\,d\upsilon'\right]. \tag{6.81}$$

通过使方程两边方括号内的数量相等：(6.81) 并取消公共 $e^{j\omega t}$ 因子，我们得到相量域表达式

$$\widetilde{V}(\mathbf{R}) = \frac{1}{4\pi\varepsilon}\int_{\upsilon'}\frac{\tilde{\rho}_{\mathrm{v}}(\mathbf{R}_{\mathrm{i}})\,e^{-jkR'}}{R'}\,d\upsilon' \quad (\mathrm{V}). \tag{6.82}$$

对于任何给定的电荷分布，方程。(6.82) 可用于计算 ~V (R)。然后得到的表达式可以在等式中使用。(6.80) 求 V (R,t)。类似地，A(R,t) 的表达式由方程式给出：(6.75) 可以转化为

$$\mathbf{A}(\mathbf{R},t) = \mathfrak{Re}\left[\widetilde{\mathbf{A}}(\mathbf{R})\,e^{j\omega t}\right] \tag{6.83}$$

和

$$\widetilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi}\int_{\upsilon'}\frac{\widetilde{\mathbf{J}}(\mathbf{R}_{\mathrm{i}})\,e^{-jkR'}}{R'}\,d\upsilon', \tag{6.84}$$

其中~J(R_i)是对应于J(R_i,t)的相量函数。对应于 ~A 的磁场相量 ~H 由下式给出

$$\widetilde{\mathbf{H}} = \frac{1}{\mu}\nabla\times\widetilde{\mathbf{A}}. \tag{6.85}$$

回想一下，时域中的微分相当于相量域中和非导电介质 (J = 0) 中乘以 jω，因此安培定律 由方程 1 给出。(6.41) 变为

$$\nabla\times\widetilde{\mathbf{H}} = j\omega\varepsilon\widetilde{\mathbf{E}} \quad \text{or} \quad \widetilde{\mathbf{E}} = \frac{1}{j\omega\varepsilon}\nabla\times\widetilde{\mathbf{H}}. \tag{6.86}$$

因此，给定具有相量 ~J 的时谐波电流密度分布，方程：(6.84)至(6.86)可依次使用

确定 ~E 和 ~H。相量向量 ~E 和 ~H 也通过法拉第定律的相量形式相关：

$$\nabla\times\widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}} \quad \text{or} \quad \widetilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla\times\widetilde{\mathbf{E}}. \tag{6.87}$$

例 6-8：将 E 与 H 联系起来

在具有 $\varepsilon = 16\varepsilon_0$ 和 $\mu = \mu_0$ 的非导电介质中，电磁波的电场强度为

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,10\sin(10^{10}t - kz) \quad (\mathrm{V/m}). \tag{6.88}$$

确定相关的磁场强度 H 并求出 k 值。

解：我们首先找到 E(z,t) 的相量 ~E(z)。由于 E(z,t) 是作为正弦函数给出的，而本书中相量是参考余弦函数定义的，因此我们重写方程：(6.88) 作为

$$\begin{aligned}\mathbf{E}(z,t) &= \hat{\mathbf{x}}\,10\cos(10^{10}t - kz - \pi/2) \quad (\mathrm{V/m})\\ &= \mathfrak{Re}\left[\widetilde{\mathbf{E}}(z)\,e^{j\omega t}\right],\end{aligned} \tag{6.89}$$

$\omega = 10^{10}$ (rad/s) 和

$$\begin{aligned}\widetilde{\mathbf{E}}(z) &= \hat{\mathbf{x}}\,10e^{-jkz}e^{-j\pi/2}\\ &= -\hat{\mathbf{x}}j10e^{-jkz}.\end{aligned} \tag{6.90}$$

为了找到 ~H(z) 和 k，我们将执行一个 "循环" ：我们将使用法拉第定律中给定的 ~E(z) 表达式来找到 ~H(z)；然后我们将使用安培定律 中的~H(z)来找到~E(z)，然后将其与~E(z)的原始表达式进行比较；比较将得出 k 的值。方程的应用(6.87) 给出

$$\begin{aligned}\widetilde{\mathbf{H}}(z) &= -\frac{1}{j\omega\mu}\nabla\times\widetilde{\mathbf{E}}\\ &= -\frac{1}{j\omega\mu}\begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ \partial/\partial x & \partial/\partial y & \partial/\partial z\\ -j10e^{-jkz} & 0 & 0\end{vmatrix}\\ &= -\frac{1}{j\omega\mu}\left[\hat{\mathbf{y}}\,\frac{\partial}{\partial z}(-j10e^{-jkz})\right]\\ &= -\hat{\mathbf{y}}j\,\frac{10k}{\omega\mu}e^{-jkz}.\end{aligned} \tag{6.91}$$

So far, we have used Eq. (6.90) for $\widetilde{\mathbf{E}}(z)$ to find $\widetilde{\mathbf{H}}(z)$, but k remains unknown. To find k, we use $\widetilde{\mathbf{H}}(z)$ in Eq. (6.86) to find $\widetilde{\mathbf{E}}(z)$:

$$\begin{aligned}\widetilde{\mathbf{E}}(z) &= \frac{1}{j\omega\varepsilon}\nabla\times\widetilde{\mathbf{H}} \\ &= \frac{1}{j\omega\varepsilon}\left[-\hat{\mathbf{x}}\frac{\partial}{\partial z}\left(-j\frac{10k}{\omega\mu}e^{-jkz}\right)\right] \\ &= -\hat{\mathbf{x}}j\frac{10k^2}{\omega^2\mu\varepsilon}e^{-jkz}. \end{aligned} \tag{6.92}$$

Equating Eqs. (6.90) and (6.92) leads to

$$k^2 = \omega^2\mu\varepsilon,$$

or

$$\begin{aligned}k &= \omega\sqrt{\mu\varepsilon} \\ &= 4\omega\sqrt{\mu_0\varepsilon_0} \\ &= \frac{4\omega}{c} = \frac{4\times10^{10}}{3\times10^8} = 133 \quad \text{(rad/m)}. \end{aligned} \tag{6.93}$$

With k known, the instantaneous magnetic field intensity is then given by

$$\begin{aligned}\mathbf{H}(z,t) &= \mathfrak{Re}\left[\widetilde{\mathbf{H}}(z)\,e^{j\omega t}\right] \\ &= \mathfrak{Re}\left[-\hat{\mathbf{y}}j\frac{10k}{\omega\mu}e^{-jkz}e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}\,0.11\sin(10^{10}t-133z) \quad \text{(A/m)}. \end{aligned} \tag{6.94}$$

We note that k has the same expression as the phase constant of a lossless transmission line [Eq. (2.49)].

Exercise 6-7: The magnetic field intensity of an electromagnetic wave propagating in a lossless medium with $\varepsilon = 9\varepsilon_0$ and $\mu = \mu_0$ is

$$\mathbf{H}(z,t) = \hat{\mathbf{x}}\,0.3\cos(10^8t - kz + \pi/4) \quad \text{(A/m)}.$$

Find $\mathbf{E}(z,t)$ and k.

Answer: $\mathbf{E}(z,t) = -\hat{\mathbf{y}}\,37.7\cos(10^8t - z + \pi/4)$ (V/m); $k = 1$ (rad/m). (See EM.)

Chapter 6 Summary

Concepts

- Faraday's law states that a voltage is induced across the terminals of a loop if the magnetic flux linking its surface changes with time.
- In an ideal transformer, the ratios of the primary to secondary voltages, currents, and impedances are governed by the turns ratio.
- Displacement current accounts for the "apparent" flow of charges through a dielectric. In reality, charges of opposite polarity accumulate along the two ends of a dielectric, giving the appearance of current flow through it.
- Boundary conditions for the electromagnetic fields at the interface between two different media are the same for both static and dynamic conditions.
- The charge continuity equation is a mathematical statement of the law of conservation of electric charge.
- Excess charges in the interior of a good conductor dissipate very quickly; through a rearrangement process, the excess charge is transferred to the surface of the conductor.
- In the dynamic case, the electric field **E** is related to both the scalar electric potential V and the magnetic vector potential **A**.
- The retarded scalar and vector potentials at a given observation point take into account the finite time required for propagation between their sources, the charge and current distributions, and the location of the observation point.

到目前为止，我们已经使用了等式。 (6.90) 让 ˜E(z) 找到 ˜H(z)，但 k 仍然未知。为了找到 k，我们在等式中使用 ˜H(z)。 (6.86) 求 ˜E(z)：

$$\begin{aligned}\widetilde{\mathbf{E}}(z) &= \frac{1}{j\omega\varepsilon}\nabla\times\widetilde{\mathbf{H}} \\ &= \frac{1}{j\omega\varepsilon}\left[-\hat{\mathbf{x}}\frac{\partial}{\partial z}\left(-j\frac{10k}{\omega\mu}e^{-jkz}\right)\right] \\ &= -\hat{\mathbf{x}}j\frac{10k^2}{\omega^2\mu\varepsilon}e^{-jkz}. \end{aligned} \tag{6.92}$$

等式(6.90) 和 (6.92) 导致

$$k^2 = \omega^2\mu\varepsilon,$$

或者

$$\begin{aligned}k &= \omega\sqrt{\mu\varepsilon} \\ &= 4\omega\sqrt{\mu_0\varepsilon_0} \\ &= \frac{4\omega}{c} = \frac{4\times10^{10}}{3\times10^8} = 133 \quad \text{(rad/m)}. \end{aligned} \tag{6.93}$$

当 k 已知时，瞬时磁场强度由下式给出

$$\begin{aligned}\mathbf{H}(z,t) &= \mathfrak{Re}\left[\widetilde{\mathbf{H}}(z)\,e^{j\omega t}\right] \\ &= \mathfrak{Re}\left[-\hat{\mathbf{y}}j\frac{10k}{\omega\mu}e^{-jkz}e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}\,0.11\sin(10^{10}t-133z) \quad \text{(A/m)}. \end{aligned} \tag{6.94}$$

我们注意到 k 与无损传输线的相位常数具有相同的表达式 [方程 1]。 (2.49)]。

练习 6-7：在具有 $\varepsilon = 9\varepsilon_0$ 和 $\mu = \mu_0$ 的无损介质中传播的电磁波的磁场强度为

$$\mathbf{H}(z,t) = \hat{\mathbf{x}}\,0.3\cos(10^8t - kz + \pi/4) \quad \text{(A/m)}.$$

求 E(z,t) 和 k。

答案：E(z,t) = $-\hat{\mathbf{y}}$ 37.7 cos(10^8t − z + π/4) (V/m)； k = 1 (rad/m)。 （参见EM。）

第六章总结

概念

- 法拉第定律指出，如果连接环路表面的磁通量随时间变化，则环路两端就会感应出电压。
- 在理想变压器中，初级与次级电压、电流和阻抗之比由匝数比决定。
- 位移电流是指电荷通过电介质的“表观”流动。实际上，相反极性的电荷沿着电介质的两端累积，呈现出电流流过电介质的现象。
- 两种不同介质之间界面处的电磁场边界条件在静态和动态条件下都是相同的。
- 电荷连续性方程是电荷守恒定律的数学表述。
- 良导体内部的多余电荷很快就会消散。通过重新排列过程，多余的电荷被转移到导体的表面。
- 在动态情况下，电场 E 与标量电势 V 和磁矢量势 A 相关。
- 给定观察点处的延迟标量和矢量势考虑了它们的源之间传播所需的有限时间、电荷和电流分布以及观察点的位置。